



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

Quasi Periodic Eruptions

ACTIVE GALACTIC NUCLEI

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Characteristics

- $10^{42-43} \text{ erg s}^{-1}$ Peak Luminosities
- Soft X-Ray Spectrum Without Hard Coronal X-Rays
- Hours/Days Recurrence Times
- $kT \sim 50 - 80 \text{ eV}$ Accretion Disk Quiescent Emission
- $kT \sim 100 - 250 \text{ eV}$ Thermal Like Flares
- Expanding Emitting Region Temperature Hysteresis
- Sometimes Regular Double Pattern
- $10^{5.5-6.5} M_{\odot}$ Mass Systems

Characteristics

- Strong TDE/CLAGN Connection
- Post Starburst Galaxy Overrepresentation
- Missing Optical Broad Lines (No BLR)
- Narrow Lines Present (Past AGN Like Activity)
- Extended Emission Line Regions (Short Lived SMBH Flare Ionization)

Models

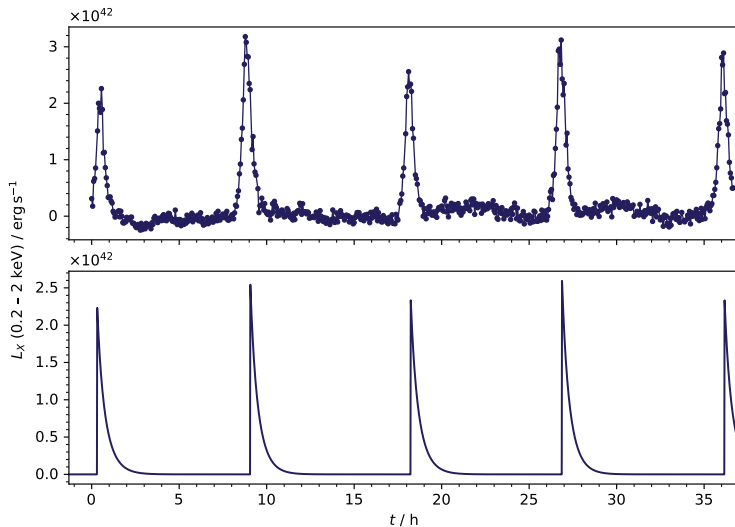
■ Disk Instabilities

- Accretion Flow Ring Oscillation (Stable/Unstable Mode Cooling/Heating)
- Warped Accretion Disks (Below Viscous Timescale Discrete Ring Breakup)

■ Orbiter Interactions

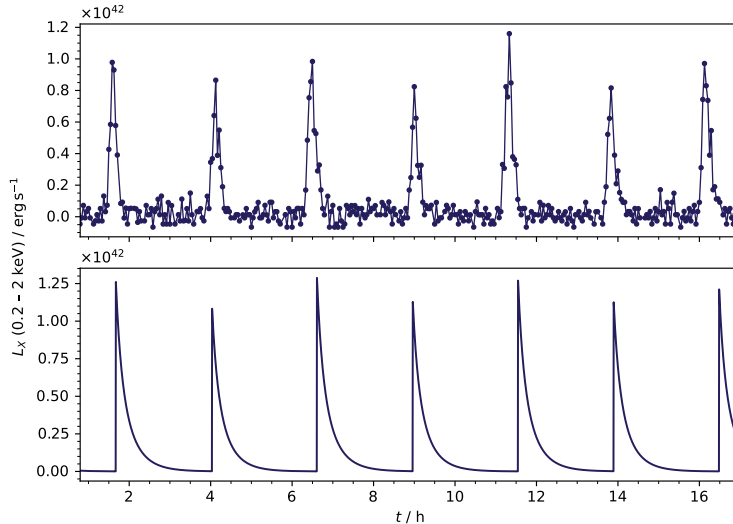
- Repeating Partial Tidal Disruptions (pTDEs)
- Roche Lobe Overflow ($\epsilon \ll 1$ MS Star)
- Extreme Mass Ratio Inspiral (SMO/BH EMRI)

Source: GSN-069



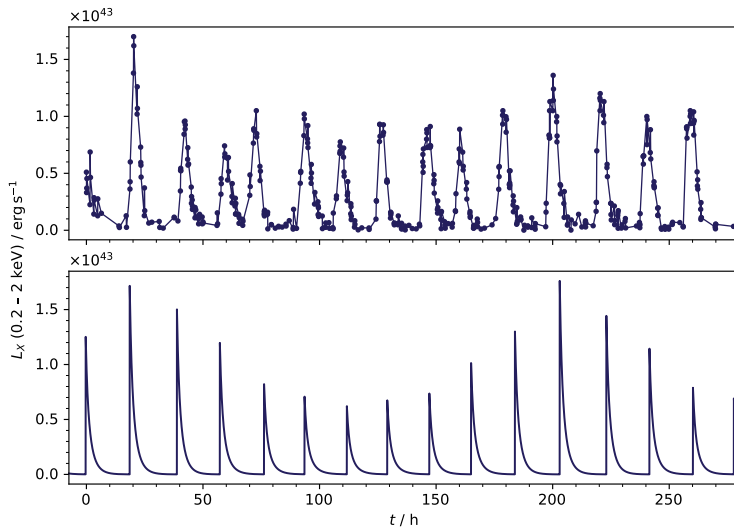
Examples

Source: eRO-QPE2



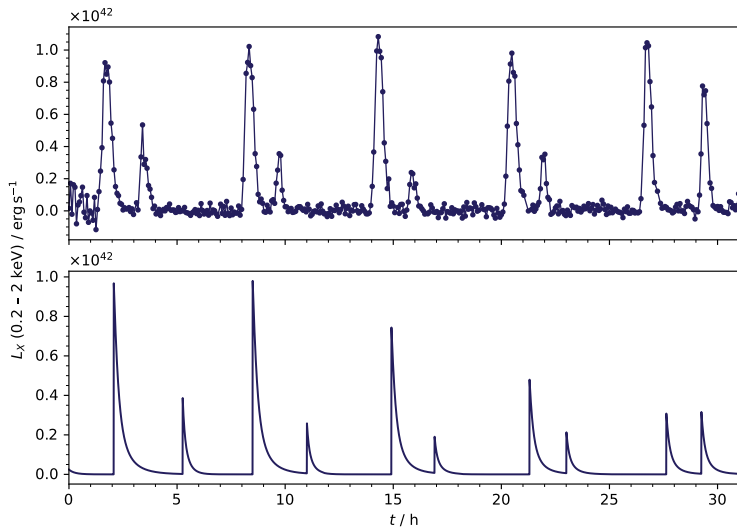
Examples

Source: eRO-QPE1



Examples

Source: RX-J1302



Examples

Outlook

- Longer Term XMM-Newton / NICER Observations (Exclude Models)
- LISA Gravitational Wave Detection (EMRI Candidates)
- Galactic Nuclei Stellar Dynamics
- Detailed Accretion Disk Structure
- Precision Cosmology Measurements

Thank You!

- E. Kara & J. García, *Supermassive Black Holes in X-Rays: Standard Accretion to Extreme Transients* (2025) arXiv:2503.22791 [astro-ph.HE]
- A. Franchini et. al., *Quasi Periodic Eruptions from Impacts with Rigidly Precessing Accretion Disc in Extreme Mass Ratio Inspiral Systems* (2023) A&A Volume 675 Article 100

Resources on GitHub: [here](#)